SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – CONCEPT MAPPING

Course Code:	CSA65	**Course Title:**	Generative AI and Large Language Models
CO Assessed:	CO1 – Precision (BL1, BL2)	**Assessment:**	Assessment Tool 1 – Concept Mapping
Weightage:	20% of CIA	**Total Marks:**	100 (2 Questions × 50 Marks)
CO1 Statement:	*Explain the fundamentals of Artificial Intelligence, Generative AI, Foundation Models, Transformer architecture, tokenization, embeddings, and Large Language Models. (BL1, BL2)*		
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Section / Batch:		**Date:**	

Instructions to Students

- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
- Draw a clear and neat concept map for each question.
- Identify all key concepts and arrange them in a logical hierarchy.
- Show meaningful linkages between concepts using arrows and connecting words.
- Compare related concepts wherever required and justify the relationships clearly.
- Ensure the concept map is well-organized, readable, and properly labelled.

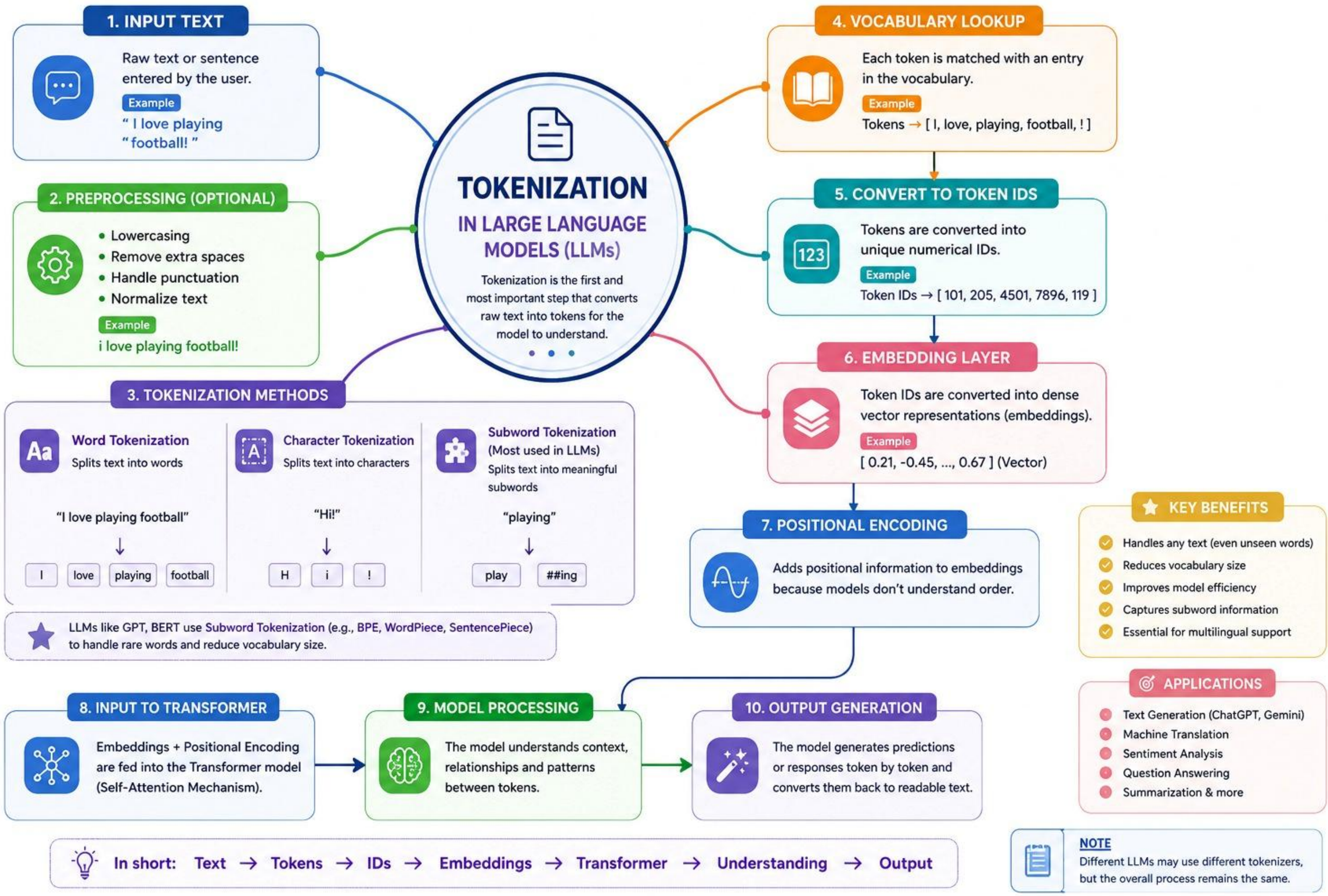
Code of Conduct

I ___________________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

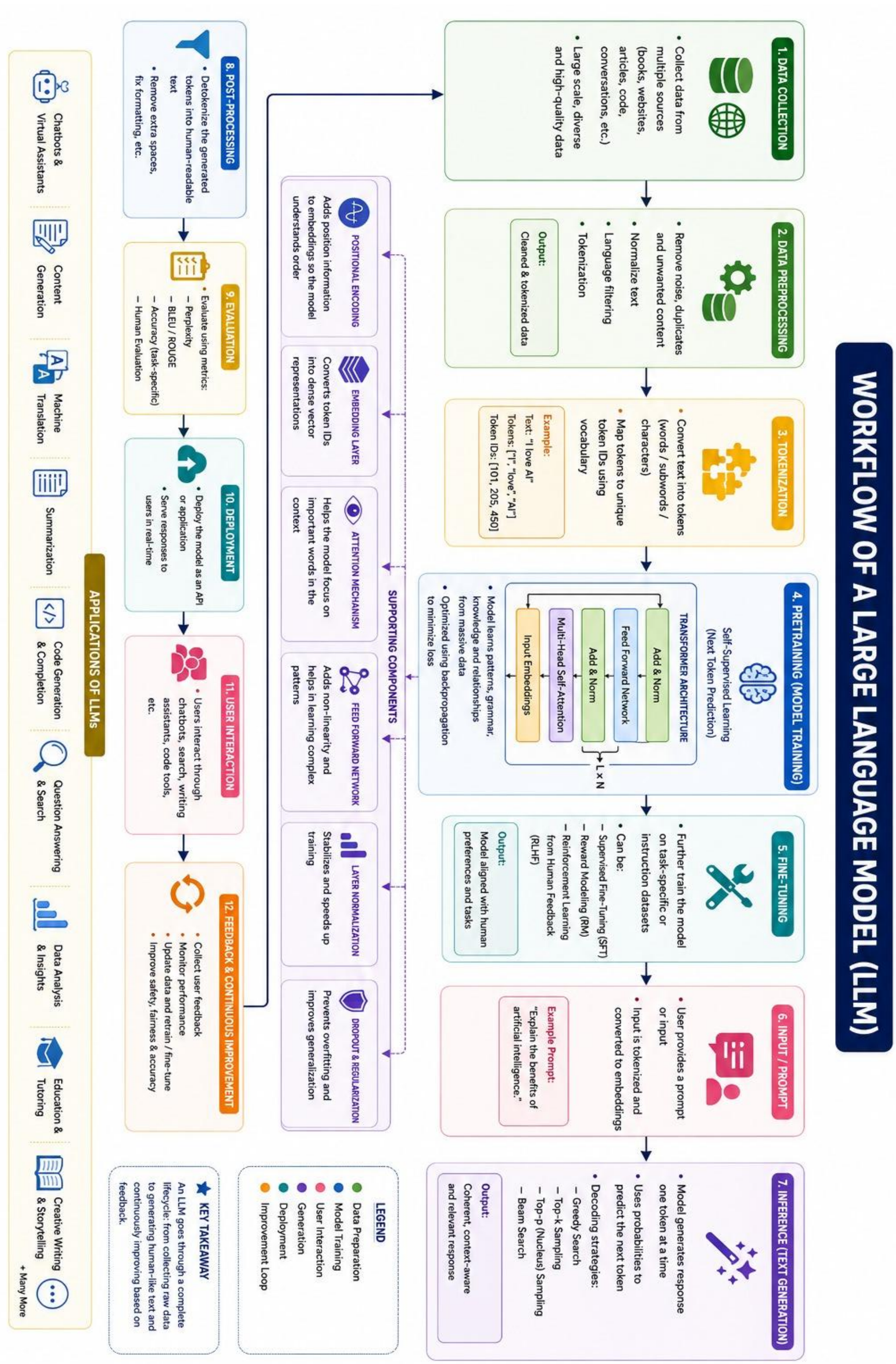
Signature of the Student: __________________________

SECTION A – CONCEPT MAPPING

9. Construct a concept map explaining the Tokenization process in Large Language Models.


TOKENIZATION
IN LARGE LANGUAGE MODELS (LLMs)
Tokenization is the first and most important step that converts raw text into tokens for the model to understand.
1. INPUT TEXT
Raw text or sentence entered by the user.
Example
" I love playing " football! "
2. PREPROCESSING (OPTIONAL)
Lowercasing
Remove extra spaces
Handle punctuation
Normalize text
Example
i love playing football!
3. TOKENIZATION METHODS
Word Tokenization
Splits text into words
"I love playing football"
I
love
playing
football
Character Tokenization
Splits text into characters
"Hi!"
H
i
!
Subword Tokenization
(Most used in LLMs)
Splits text into meaningful subwords
"playing"
play
##ing
LLMs like GPT, BERT use Subword Tokenization (e.g., BPE, WordPiece, SentencePiece) to handle rare words and reduce vocabulary size.
4. VOCABULARY LOOKUP
Each token is matched with an entry in the vocabulary.
Example
Tokens → [I, love, playing, football, !]
5. CONVERT TO TOKEN IDS
Tokens are converted into unique numerical IDs.
Example
Token IDs → [101, 205, 4501, 7896, 119]
6. EMBEDDING LAYER
Token IDs are converted into dense vector representations (embeddings).
Example
[0.21, -0.45, ..., 0.67] (Vector)
7. POSITIONAL ENCODING
Adds positional information to embeddings because models don't understand order.
KEY BENEFITS
Handles any text (even unseen words)
Reduces vocabulary size
Improves model efficiency
Captures subword information
Essential for multilingual support
APPLICATIONS
Text Generation (ChatGPT, Gemini)
Machine Translation
Sentiment Analysis
Question Answering
Summarization & more
8. INPUT TO TRANSFORMER
Embeddings + Positional Encoding are fed into the Transformer model (Self-Attention Mechanism).
9. MODEL PROCESSING
The model understands context, relationships and patterns between tokens.
10. OUTPUT GENERATION
The model generates predictions or responses token by token and converts them back to readable text.
In short: Text → Tokens → IDs → Embeddings → Transformer → Understanding → Output
NOTE
Different LLMs may use different tokenizers, but the overall process remains the same.

32. Draw a concept map showing the complete workflow of a Large Language Model.



RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical / Concept	30%	All key concepts (AI, ML, DL, GenAI, LLMs) correctly identified with clear hierarchical relationships	Most concepts correct; minor gaps in relationships	Several concepts missing or misclassified	Major conceptual errors; map incomplete
Application	25%	Real-world examples linked accurately to each concept	Examples mostly relevant	Few or generic examples	No real-world linkage shown
Quality	20%	Logical, well-organized structure with clear connectors	Mostly organized; minor structural issues	Some disorganization affecting clarity	Poorly structured, hard to follow
Presentation	15%	Visually clear, creative, easy to interpret	Neat and readable	Cluttered or inconsistent formatting	Untidy, difficult to interpret
Documentation / Communication	10%	Concise labels/notes that add clarity	Adequate labeling	Minimal or unclear labeling	No supporting labels or explanation

Marks Evaluation:

Question No.	Technical / Concept(30%)	Application(25%)	Quality (20%)	Presentation (15%)	Documentation / Communication (10%)	Total Marks (Each – 50)
Q1						
Q2						
					Overall Total (100)	

Faculty In-charge	Course Coordinator	Head of Department
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________